

ENGR 225: Mechanics 1

Homework Coversheet

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ASSIGNMENT: Homework 1

Level of involvement of AI, LLMs, or related tools in the completion of this assignment:

- ☐ None
- ☒ Minor
- ☐ Major
- ☐ Extensive

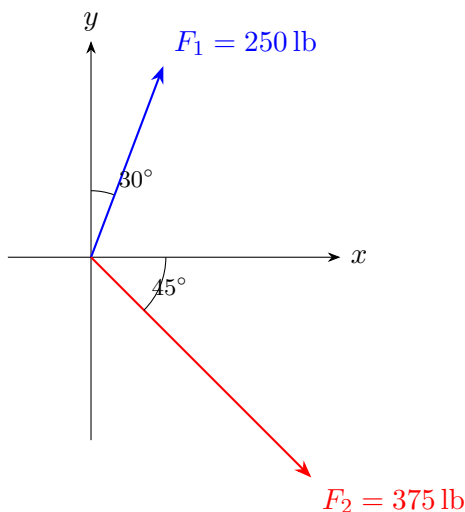
Mark the box that applies before submitting. Your disclosure does not affect your grade – it keeps you accountable for the work you are doing.

ENGR 225 – Homework 1 – Solutions

Problem 1

Given / Find: $F_1 = 250$ lb acting 30° from the $+y$ -axis (tilted toward $+x$); $F_2 = 375$ lb acting 45° below the $+x$ -axis. Find the magnitude of the resultant $\vec{F}_R = \vec{F}_1 + \vec{F}_2$ and its direction θ , measured counterclockwise from the $+x$ -axis.

Sketch:



Equations & Solution:

Step 1 – Resolve each force into x, y components. Since F_1 is measured from the y -axis, its horizontal component uses sine and its vertical component uses cosine:

$$F_{1x} = F_1 \sin 30^\circ = (250)(0.5) = 125 \text{ lb}$$

$$F_{1y} = F_1 \cos 30^\circ = (250)(0.8660) = 216.5 \text{ lb}$$

F_2 is measured from the x -axis, below it, so its y -component is negative:

$$F_{2x} = F_2 \cos 45^\circ = (375)(0.7071) = 265.2 \text{ lb}$$

$$F_{2y} = -F_2 \sin 45^\circ = -(375)(0.7071) = -265.2 \text{ lb}$$

Step 2 – Sum components.

$$R_x = F_{1x} + F_{2x} = 125 + 265.2 = 390.2 \text{ lb}$$

$$R_y = F_{1y} + F_{2y} = 216.5 - 265.2 = -48.7 \text{ lb}$$

Step 3 – Magnitude and direction.

$$F_R = \sqrt{R_x^2 + R_y^2} = \sqrt{(390.2)^2 + (48.7)^2} = 393 \text{ lb}$$

$$\theta' = \tan^{-1}\left(\frac{48.7}{390.2}\right) = 7.11^\circ \quad (\text{below } +x\text{-axis, since } R_y < 0)$$

Measuring counterclockwise from the $+x$ -axis all the way around to the resultant:

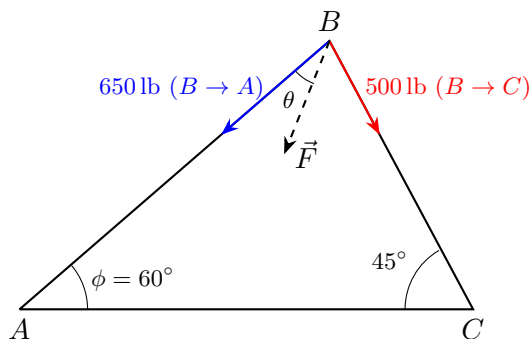
$$\theta = 360^\circ - 7.11^\circ = 352.9^\circ \approx 353^\circ$$

Answer
$F_R = 393 \text{ lb}, \quad \theta = 353^\circ \text{ measured counterclockwise from the } +x\text{-axis.}$

Problem 2

Given / Find: Force $\vec{F}$ acts on frame ABC such that its component along member AB is 650 lb (directed $B \rightarrow A$) and its component along BC is 500 lb (directed $B \rightarrow C$), with $\phi = 60^\circ$ at A and 45° at C . Find F and its direction θ (measured from the AB axis at B).

Sketch:



Equations & Solution:

Step 1 – Interior angle at B . The triangle's angles sum to 180° :

$$\angle B = 180^\circ - \phi - 45^\circ = 180^\circ - 60^\circ - 45^\circ = 75^\circ$$

This is the angle between the given 650-lb and 500-lb components.

Step 2 – Resultant magnitude. The 650-lb and 500-lb components add to give $\vec{F}$:

$$\begin{aligned} F &= \sqrt{(650)^2 + (500)^2 + 2(650)(500) \cos 75^\circ} \\ &= \sqrt{422500 + 250000 + 168232} = \sqrt{840732} \\ F &= 917 \text{ lb} \end{aligned}$$

Step 3 – Direction. In the force triangle, the interior angle opposite F is $180^\circ - 75^\circ = 105^\circ$. Solving for the angle θ between F and the 650-lb (AB) direction:

$$\begin{aligned} \frac{500}{\sin \theta} &= \frac{917}{\sin 105^\circ} \\ \sin \theta &= \frac{500 \sin 105^\circ}{917} = \frac{482.9}{917} = 0.5267 \\ \theta &= 31.8^\circ \end{aligned}$$

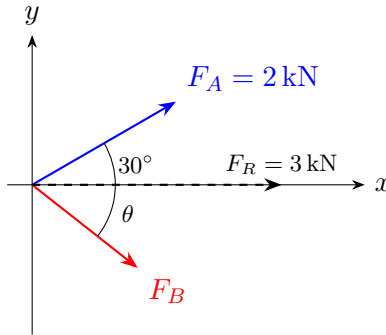
Answer

$F = 917 \text{ lb}$, $\theta = 31.8^\circ$ from the AB axis.

Problem 3

Given / Find: Two tugboats pull a barge: $\vec{F}_A = 2 \text{ kN}$ at 30° above the $+x$ -axis, and $\vec{F}_B$ (unknown) below the x -axis at angle θ . The resultant is 3 kN directed along the $+x$ -axis. Find F_B and θ .

Sketch:



Equations & Solution:

Step 1 – Components of F_A .

$$F_{Ax} = 2 \cos 30^\circ = 1.732 \text{ kN}$$

$$F_{Ay} = 2 \sin 30^\circ = 1.000 \text{ kN}$$

Step 2 – Require the resultant equal 3 kN along $+x$ with zero y -component.

$$F_{Ax} + F_{Bx} = 3 \quad \Rightarrow \quad F_{Bx} = 3 - 1.732 = 1.268 \text{ kN}$$

$$F_{Ay} + F_{By} = 0 \quad \Rightarrow \quad F_{By} = -1.000 \text{ kN (below } x\text{-axis)}$$

Step 3 – Magnitude and direction of F_B .

$$F_B = \sqrt{(1.268)^2 + (1.000)^2} = \sqrt{1.608 + 1.000} = 1.61 \text{ kN}$$

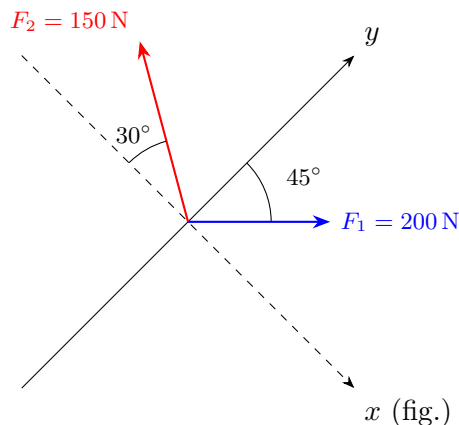
$$\theta = \tan^{-1}\left(\frac{1.000}{1.268}\right) = 38.2^\circ \text{ below the } +x\text{-axis}$$

Answer
$F_B = 1.61 \text{ kN}, \quad \theta = 38.2^\circ \text{ below the } +x\text{-axis.}$

Problem 4

Given / Find: $F_1 = 200\text{ N}$ acts horizontally. The y -axis is 45° from horizontal (from F_1), and the figure's x -axis (the dashed reference line) is 90° from the y -axis — i.e. also 45° from F_1 , on the other side, as shown. $F_2 = 150\text{ N}$ acts 30° from this x -axis reference line. Find the resultant magnitude and its direction θ , measured counterclockwise from the positive x -axis defined in the figure.

Sketch:



Equations & Solution:

Step 1 – Locate each force relative to the figure's x -axis. F_1 is horizontal, with the y -axis 45° above it; since the figure's x -axis is 90° from y , it falls 45° below F_1 on the other side, so F_1 makes a 45° angle with the figure's x -axis. F_2 is 30° from the same x -axis line, on the side toward the y -axis — i.e. 30° short of the line's upper ray — so measured from the figure's positive x -axis it lies at $180^\circ - 30^\circ = 150^\circ$.

Step 2 – Components (in the figure's x - y system).

$$F_{1x} = 200 \cos 45^\circ = 141.4\text{ N}$$

$$F_{1y} = 200 \sin 45^\circ = 141.4\text{ N}$$

$$F_{2x} = 150 \cos 150^\circ = -129.9\text{ N}$$

$$F_{2y} = 150 \sin 150^\circ = 75.0\text{ N}$$

Step 3 – Sum and resolve.

$$R_x = 141.4 - 129.9 = 11.5\text{ N}$$

$$R_y = 141.4 + 75.0 = 216.4\text{ N}$$

$$F_R = \sqrt{(11.5)^2 + (216.4)^2} = 217\text{ N}$$

$$\theta = \tan^{-1}\left(\frac{216.4}{11.5}\right) = 87.0^\circ$$

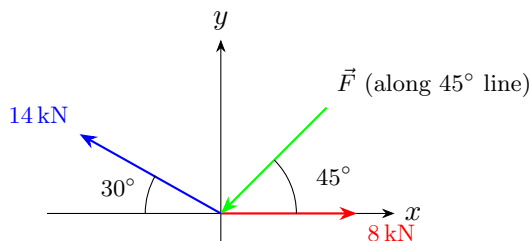
Answer

$$F_R = 217\text{ N}, \quad \theta = 87.0^\circ \text{ ccw from the figure's } +x\text{-axis.}$$

Problem 5

Given / Find: A bracket carries 14 kN at 30° above the $-x$ -axis and 8 kN along $+x$, plus a variable force $\vec{F}$ acting along a fixed 45° line. Find F so that the resultant is as small as possible, and find that minimum resultant F_R .

Sketch:



Equations & Solution:

Step 1 – Sum the two fixed forces.

$$S_x = 14 \cos 150^\circ + 8 = -12.12 + 8 = -4.12 \text{ kN}$$

$$S_y = 14 \sin 150^\circ = 7.00 \text{ kN}$$

Step 2 – Add the variable force F along the 45° line.

$$R_x = S_x + F \cos 45^\circ, \quad R_y = S_y + F \sin 45^\circ$$

$$F_R^2 = (S_x + F \cos 45^\circ)^2 + (S_y + F \sin 45^\circ)^2$$

Step 3 – Minimize: set $\frac{d(F_R^2)}{dF} = 0$.

$$\cos 45^\circ (S_x + F \cos 45^\circ) + \sin 45^\circ (S_y + F \sin 45^\circ) = 0$$

Since $\cos^2 45^\circ + \sin^2 45^\circ = 1$:

$$F = -(S_x \cos 45^\circ + S_y \sin 45^\circ) = -[(-4.12)(0.7071) + (7.00)(0.7071)] = -2.03 \text{ kN}$$

$$F = 2.03 \text{ kN}$$

Step 4 – Compute the resulting minimum F_R .

$$R_x = -4.12 + (-2.03)(0.7071) = -5.56 \text{ kN}$$

$$R_y = 7.00 + (-2.03)(0.7071) = 5.56 \text{ kN}$$

$$F_R = \sqrt{(5.56)^2 + (5.56)^2} = 7.87 \text{ kN}$$

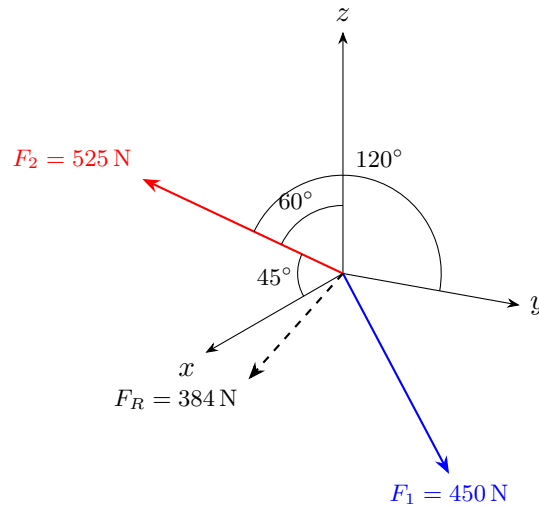
Answer

$F = 2.03 \text{ kN}$ (directed opposite the drawn 45° ray), giving a minimum resultant $F_R = 7.87 \text{ kN}$.

Problem 6

Given / Find: $F_1 = 450 \text{ N}$ lies in the y - z plane, defined by a 3-4-5 triangle ($F_{1x} = 0$). $F_2 = 525 \text{ N}$ has coordinate direction angles $\alpha = 45^\circ$ (from x), $\beta = 120^\circ$ (from y), $\gamma = 60^\circ$ (from z). Find the magnitude and coordinate direction angles of the resultant, and sketch it.

Sketch:



Equations & Solution:

Step 1 – Components of F_1 (3-4-5 triangle, in the y - z plane).

$$F_{1x} = 0$$

$$F_{1y} = 450 \left(\frac{3}{5} \right) = 270 \text{ N}$$

$$F_{1z} = -450 \left(\frac{4}{5} \right) = -360 \text{ N}$$

Step 2 – Components of F_2 (direction cosines).

$$F_{2x} = 525 \cos 45^\circ = 371.2 \text{ N}$$

$$F_{2y} = 525 \cos 120^\circ = -262.5 \text{ N}$$

$$F_{2z} = 525 \cos 60^\circ = 262.5 \text{ N}$$

$$\text{Check: } \cos^2 45^\circ + \cos^2 120^\circ + \cos^2 60^\circ = 0.5 + 0.25 + 0.25 = 1 \checkmark$$

Step 3 – Sum components.

$$R_x = 0 + 371.2 = 371.2 \text{ N}$$

$$R_y = 270 - 262.5 = 7.5 \text{ N}$$

$$R_z = -360 + 262.5 = -97.5 \text{ N}$$

Step 4 – Magnitude and coordinate direction angles.

$$F_R = \sqrt{(371.2)^2 + (7.5)^2 + (-97.5)^2} = \sqrt{147,373} = 384 \text{ N}$$

$$\alpha = \cos^{-1}\left(\frac{371.2}{384}\right) = 14.8^\circ$$

$$\beta = \cos^{-1}\left(\frac{7.5}{384}\right) = 88.9^\circ$$

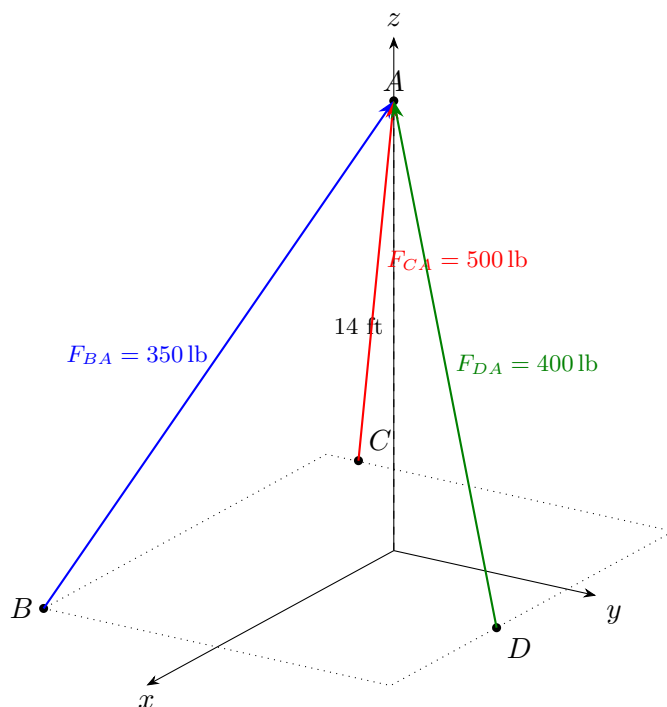
$$\gamma = \cos^{-1}\left(\frac{-97.5}{384}\right) = 105^\circ$$

Answer
$F_R = 384 \text{ N}, \quad \alpha = 14.8^\circ, \quad \beta = 88.9^\circ, \quad \gamma = 105^\circ.$

Problem 7

Given / Find: A plate is suspended from point A by three cables to corners B , C , D , with tensions $F_{BA} = 350$ lb, $F_{CA} = 500$ lb, $F_{DA} = 400$ lb. A is 14 ft above the plate. Using the in-plane offsets given in the figure, express each cable tension as a Cartesian vector.

Sketch:



Equations & Solution:

Step 1 – Coordinates. Placing the ground point directly beneath A at the origin, with A at height 14 ft, the given in-plane offsets locate the corners at:

$$A = (0, 0, 14), \quad B = (5, -6, 0), \quad C = (-3, -3, 0), \quad D = (2, 6, 0) \text{ [ft]}$$

Step 2 – Position vectors and cable lengths.

$$\vec{r}_{BA} = A - B = (-5, 6, 14) \text{ ft},$$

$$L_{BA} = \sqrt{5^2 + 6^2 + 14^2} = 16.03 \text{ ft}$$

$$\vec{r}_{CA} = A - C = (3, 3, 14) \text{ ft},$$

$$L_{CA} = \sqrt{3^2 + 3^2 + 14^2} = 14.63 \text{ ft}$$

$$\vec{r}_{DA} = A - D = (-2, -6, 14) \text{ ft},$$

$$L_{DA} = \sqrt{2^2 + 6^2 + 14^2} = 15.36 \text{ ft}$$

Step 3 – Unit vectors and force vectors ($\vec{F} = F \cdot \hat{u} = F \vec{r}/L$).

$$\vec{F}_{BA} = 350 \cdot \frac{(-5, 6, 14)}{16.03} = -109\hat{i} + 131\hat{j} + 306\hat{k} \text{ lb}$$

$$\vec{F}_{CA} = 500 \cdot \frac{(3, 3, 14)}{14.63} = 103\hat{i} + 103\hat{j} + 479\hat{k} \text{ lb}$$

$$\vec{F}_{DA} = 400 \cdot \frac{(-2, -6, 14)}{15.36} = -52.1\hat{i} - 156\hat{j} + 365\hat{k} \text{ lb}$$

Answer

$$\vec{F}_{BA} = -109\hat{i} + 131\hat{j} + 306\hat{k} \text{ lb}$$

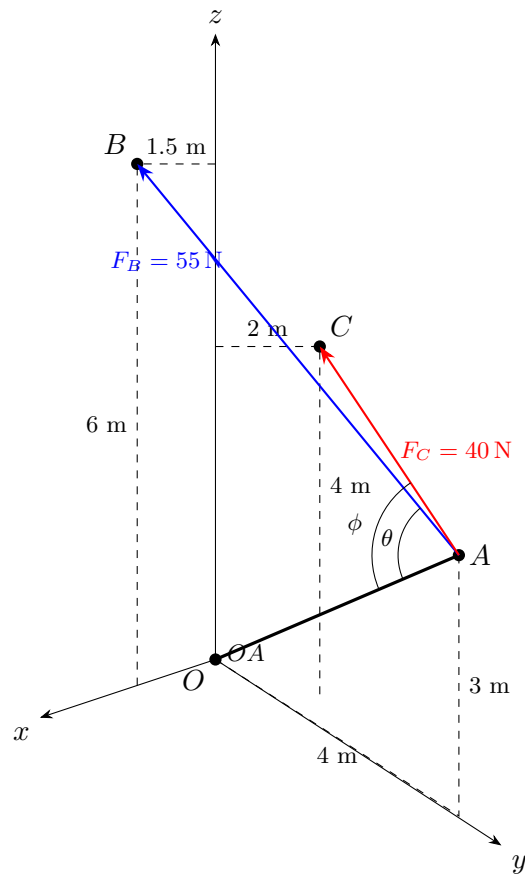
$$\vec{F}_{CA} = 103\hat{i} + 103\hat{j} + 479\hat{k} \text{ lb}$$

$$\vec{F}_{DA} = -52.1\hat{i} - 156\hat{j} + 365\hat{k} \text{ lb}$$

Problem 8

Given / Find: Flagpole OA (O at the base, A at the top) is connected by cables AB and AC anchored to a wall. Using the dimensions shown, find the angles θ (between OA and AB) and ϕ (between OA and AC).

Sketch:



Equations & Solution:

Step 1 – Coordinates from the given dimensions.

$$O = (0, 0, 0), \quad A = (0, 4, 3), \quad B = (1.5, 0, 6), \quad C = (-2, 0, 4) \text{ [m]}$$

Step 2 – Vectors from vertex A .

$$\vec{AO} = O - A = (0, -4, -3) \text{ m},$$

$$|\vec{AO}| = \sqrt{4^2 + 3^2} = 5.00 \text{ m}$$

$$\vec{AB} = B - A = (1.5, -4, 3) \text{ m},$$

$$|\vec{AB}| = \sqrt{1.5^2 + 4^2 + 3^2} = 5.22 \text{ m}$$

$$\vec{AC} = C - A = (-2, -4, 1) \text{ m},$$

$$|\vec{AC}| = \sqrt{2^2 + 4^2 + 1^2} = 4.58 \text{ m}$$

Step 3 – Angle θ between AO and AB .

$$\begin{aligned}\vec{AO} \cdot \vec{AB} &= (0)(1.5) + (-4)(-4) + (-3)(3) = 0 + 16 - 9 = 7 \\ \cos \theta &= \frac{7}{(5.00)(5.22)} = 0.2682 \\ \theta &= 74.4^\circ\end{aligned}$$

Step 4 – Angle ϕ between AO and AC .

$$\begin{aligned}\vec{AO} \cdot \vec{AC} &= (0)(-2) + (-4)(-4) + (-3)(1) = 0 + 16 - 3 = 13 \\ \cos \phi &= \frac{13}{(5.00)(4.58)} = 0.5674 \\ \phi &= 55.4^\circ\end{aligned}$$

Answer
$\theta = 74.4^\circ, \quad \phi = 55.4^\circ.$